

PAVAN KUMAR

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Professional Summary

AI/ML Engineer with 5 years of end-to-end experience building and deploying LLM-powered applications, RAG systems, agentic AI workflows, and production ML pipelines across fintech and enterprise automation. Deep expertise in GenAI (LLaMA 2, GPT-4, RAG), MLOps (Docker, Kubernetes, MLflow), and model risk governance aligned with MRM and GDPR. Published researcher (Interspeech 2025, ACM 2024) with a track record of translating complex AI systems into measurable business outcomes.

Publications

- Yarlagadda, Rakesh & Aggarwal, Pranjal & Jamadagni, Vaibhav & Mahajani, Ghritachi & Malasani, Pavan & Nirjhar, Ehsanul Haque & Chaspari, Theodora. (2024). **An AI-Powered Interactive Interface to Enhance Accessibility of Interview Training for Military Veterans**. 82-84. 10.1145/3686215.3688371.
- Aggarwal, Pranjal, Ghritachi Mahajani, Pavan Kumar Malasani, Vaibhav Jamadagni, Caroline J. Wendt, Ehsanul Haque Nirjhar, and Theodora Chaspari. "Investigating the Reasoning Abilities of Large Language Models for Understanding Spoken Language in Interpersonal Interactions." In Proc. Interspeech 2025, pp. 4518-4522. 2025.

Technical Skills

- Statistical & ML Models:** Logistic Regression, XGBoost, LightGBM, Random Forest, SVM, CNN, RNN, Isolation Forest, Gradient Boosting, Reinforcement Learning (PPO, Q-Learning, Actor-Critic, Policy Gradient), Time-Series Forecasting, Anomaly Detection
- Programming & Big Data:** Python, SQL, PySpark, Spark, R, C++, Snowflake, AWS Redshift, ETL Pipeline Design, Apache Kafka
- ML Frameworks:** TensorFlow, PyTorch, Scikit-learn, Keras, XGBoost, LightGBM, OpenAI, Hugging Face, LangChain, MLflow
- Computer Vision & Signal Processing:** OpenCV, PyTorch Vision, Wafer Defect Detection, Image Analytics, WavLM, Whisper (Audio Processing)
- GenAI & LLM:** LLaMA 2 (LoRA Fine-tuning), GPT-4, Gemini-1.5 Pro, RAG Pipelines, Pinecone, GPTQ/AWQ Quantization, Model Context Protocol (MCP), Few-Shot Prompting
- MLOps & Deployment:** Docker, Kubernetes, Apache Airflow, FastAPI, Jenkins, Model Monitoring, AWS SageMaker, Azure ML, GCP Vertex AI
- Domain Expertise:** Large-Scale ML Data Pipelines, ETL Workflow Design, Batch Data Processing, Image & Audio Data Pipelines (developing), Distributed Data Infrastructure, ML Training Data Management
- Soft Skills:** Stakeholder Communication, Cross-Functional Collaboration, Technical Documentation, Quantitative Model Validation & Reporting

Experience

AI/ML Engineer | Remote, USA | Citigroup

June 2024 - Present

- Architected a RAG pipeline using LLaMA 2 and Pinecone vector retrieval, automating market brief generation and saving 28 hours of manual research weekly across the investment banking division.
- Engineered document ingestion and chunking workflows using LangChain and OpenAI embeddings, enabling analysts to query proprietary financial filings and regulatory documents with source-traceable Q&A.
- Deployed the assistant via FastAPI on containerized Docker/Kubernetes infrastructure, ensuring high-availability with CI/CD pipelines (MLflow) for continuous model updates.
- Architected an agentic AI layer using Model Context Protocol (MCP) on top of the RAG pipeline, enabling the assistant to autonomously plan, retrieve, and synthesize information across multiple financial sources significantly reducing analyst intervention in routine research workflows.
- Collaborated with business owners and control officers to remediate integration issues and ensure alignment with enterprise model governance standards.
- Improved loan approval accuracy by 19% by engineering credit-risk scoring pipelines with Gradient Boosting, Logistic Regression, and SHAP-based Explainable AI, fully aligned with Citi's MRM framework and internal audit requirements.
- Enhanced transaction surveillance precision by 27% by validating AML anomaly detection models integrating Isolation Forest, BERT embeddings, and Kafka streams; documented validation findings per MRM policy standards.
- Established enterprise-wide benchmarks for model reliability by conducting bias testing, audit trail standardization, and retraining cadence protocols for high-impact fraud detection and credit underwriting models.
- Acted as SME for business owners, model owners, and control officers, translating complex model outputs into actionable insights for multi-region portfolios.
- Reduced model retraining cycle time by 40% by implementing scalable ML pipelines on Kubernetes-managed Linux clusters with automated feature engineering, hyperparameter tuning, and CI/CD workflows (MLflow).

AI Research Engineer | Boulder, Colorado | University of Colorado, Boulder

Aug 2023 - June 2024

- Designed and deployed a multimodal interview feedback system leveraging GPT-3.5, GPT-4, and Gemini-1.5 Pro to evaluate veteran performance and surface actionable improvement insights, driving a 7% increase in user engagement.
- Built a few-shot GPT classification model to assess veteran interview responses with task-specific accuracy, achieving a 15% improvement in precision across 10+ structured evaluation experiments.
- Engineered an audio processing pipeline using WavLM and Whisper to extract acoustic and sentiment features, enabling advanced speech interpretation and personalized feedback generation.
- Fine-tuned LLaMA using LoRA for audio classification and applied GPTQ and AWQ quantization techniques & boosting inference speed by 30% while preserving prediction quality.
- Contributed to multimodal LLM research fusing text and audio modalities, enhancing contextual classification and enabling more personalized, task-specific model outputs.
- Published research at the ACM Conference on AI and Robotics on GenAI applications for veteran interview analysis; authored task-specific LLM prompts and led 5+ knowledge-sharing sessions on prompt engineering best practices.
- Spearheaded literature reviews, data analysis, and stakeholder reporting to support research proposals, academic publications, and cross-functional communication using advanced Excel and PowerPoint.

ML Engineer | Chennai, India | TCS

Jan 2021 - July 2023

- Reduced default probability prediction error by 19% by deploying credit risk models (Random Forest, XGBoost, LightGBM, Logistic Regression) for loan approval workflows, supporting direct underwriting decisions.
- Achieved 92% precision on fraud detection across 3M+ daily records by building transaction anomaly detection using Isolation Forest and Autoencoders, ensuring compliance with RBI and GDPR.
- Generated audit-ready validation reports with SHAP and LIME for deployed credit scoring models, providing transparent decision support for compliance auditors and business units.
- Managed high-volume data ingestion across distributed systems using PySpark, Apache Spark, and Airflow (AWS Redshift, S3), ensuring high-performance model training readiness for credit scoring.
- Built secure ETL pipelines and feature stores with data engineering and business stakeholders, optimizing ingestion pipelines for marketing analytics and financial forecasting projects.
- Designed time-series forecasting solutions for financial planning, translating complex model outputs into actionable strategies and collaborating with cross-functional teams to integrate ML models into banking products.
- Built and deployed a machine learning model to automate loan approval for a new lending scheme, reducing delinquency rates by 25% through accurate risk assessment; used Python, Scikit-learn, and XGBoost for model training and evaluation.
- Performed extensive data cleaning, statistical analysis, and feature engineering on a dataset of over 500,000 records using SQL and Pandas, improving model performance by 15% by identifying key predictors like income, credit score, and debt-to-income ratio.
- Generated actionable insights through advanced data visualization in Tableau, including trend analysis and correlation matrices, enabling stakeholders to understand applicant risk profiles and aiding in decision-making to mitigate lending risks.
- Implemented MLOps practices, including Docker containerization and CI/CD pipelines with Jenkins, streamlining model deployment and enhancing cross-functional collaboration by reducing deployment time by 30%.
- Explored data imbalances and conducted class rebalancing using techniques such as SMOTE and class weighting, improving the model's AUC-ROC score and reducing False Positives by 22%, contributing to accurate loan approvals.

Projects

WorkflowPro - AI-Powered Browser Automation Platform | Tech Stack: FastAPI, SQLAlchemy, React 19, TypeScript, GPT-4 Vision, WebSocket, JWT

- Built a natural language-to-workflow automation engine using GPT-4 Vision to parse, understand, and interact with web interfaces autonomously enabling users to describe multi-step browser tasks in plain English (e.g., "log into Notion and create a project") and have them executed end-to-end.
- Designed a real-time execution engine with WebSocket-powered visual feedback and an interactive playground for workflow generation and editing, reducing the technical barrier for building browser automations from days to minutes.
- Architected enterprise-grade security with JWT authentication and encrypted credential management, integrated comprehensive execution monitoring dashboard providing audit trails for every automated action.
- *Addressed critical business challenges:* manual documentation overhead, regression test maintenance, and inefficient onboarding - enabling teams to automate repetitive web workflows without deep technical expertise.

Education

University of Colorado Boulder

Master of Science in Data Science

Boulder, CO

GPA: 3.8/4.0

Certifications

AWS Certified Generative AI Developer - Professional [Link](#)